

THURSDAY, SEPTEMBER 17TH, 2026

CLIMATE & SUSTAINABILITY

The Rock Takes It Back

Cement's carbon comes out of the limestone rather than out of the fire, which is why no fuel can fix it — and why the concrete spends the next several centuries trying to undo it.

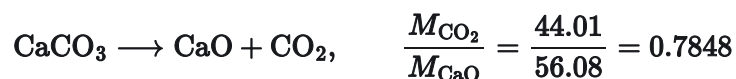
2,915 words · 15-minute read

Stand close to a Roman wall and scrape at the mortar between the bricks, and what comes away under the blade is calcite — the same mineral as the limestone in the quarry the Romans took it from. The mortar made a round trip. Someone burned limestone in a wood-fired kiln until it gave up its carbon dioxide, slaked the resulting quicklime with water, spread the paste between courses of brick, and then the paste sat in the air for two thousand years quietly reabsorbing exactly the gas the kiln had driven off. Burn, slake, carbonate, and you are back where you started. Masons have called it the lime cycle for as long as they have had a word for it, and it is one of the few genuinely closed loops in the history of construction materials.

Portland cement broke the loop, and the way it broke it is the whole subject. A modern cement does not get its strength from carbonation. It gets it from calcium silicate hydrate, a gel that forms when water attacks the alite and belite in clinker, and which is perfectly happy never to meet carbon dioxide at all. The carbon still comes out at the kiln — more of it than ever, from about 4.1 billion tonnes of cement a year — but the material has no structural need to take it back. It takes some of it back anyway, slowly, and the argument over how much is currently one of the noisier disagreements in the carbon-cycle literature, with credible published estimates differing by a factor of nearly four [6], [8], [10].

The Half That Never Sees a Flame

The reaction that matters happens before any cement exists. Limestone is calcium carbonate; clinker needs calcium oxide; the difference is a molecule of carbon dioxide per molecule of rock, and it leaves as gas.



That ratio is the entire emission factor of the cement industry, and it is not negotiable by anybody. Clinker is roughly 65 per cent lime by mass, so each tonne of it carries 0.65 tonnes of CaO that used to be carbonate, and releasing that much lime releases $0.65 \times 0.7848 = 0.510$ tonnes of carbon dioxide. That number is the default emission factor in the 2006 IPCC inventory guidelines [2], reached by exactly this arithmetic and by no measurement at all. National inventories then multiply it by 1.02 to account for kiln

dust that leaves the system already calcined, giving the 0.520 that appears in most published accounts. Run the stoichiometry the other way and a tonne of clinker needs 1.160 tonnes of calcite fed into the kiln — again a figure the guidelines quote and again one that falls straight out of the molar masses.

A THOUGHT EXPERIMENT

Suppose the kiln burned nothing at all: a perfect electric furnace on zero-carbon power, no coal, no petroleum coke, no waste tyres. How much would a tonne of clinker still emit?

Half a tonne, near enough, and that is the fact that separates cement from almost every other heavy industry. Steel can in principle be made with hydrogen; aluminium smelting is already an electrical process waiting for clean electricity; a glass furnace is a heat problem and heat can come from anywhere. Cement's emissions are chemical before they are thermal. A best-practice kiln burns about 3.4 gigajoules of fuel per tonne of clinker [5], and if that fuel were entirely carbon-free the plant would still put out roughly 0.51 tonnes of carbon dioxide per tonne of clinker, because the rock is not a fuel and the gas coming off it was never combustion products. Across the industry the split runs around 60 per cent process, 40 per cent fuel [4], [20]. Decarbonizing the energy supply addresses the smaller share.

Nine Hundred Degrees, and the Trick of Lowering It

Calcination is endothermic and reversible, which turns out to matter twice over. The standard thermochemical values give an enthalpy of about 179 kilojoules per mole of carbonate and an entropy change of about 160 joules per mole per kelvin, and if you divide one by the other you get the temperature at which the free energy vanishes: 1,117 kelvin, or 844 degrees Celsius. Every engineering handbook says 900. The gap is 55 kelvin and it is not a rounding error — it is what happens when you use room-temperature thermodynamic data a thousand degrees away from room temperature. Baker measured the decomposition pressure directly in a high-pressure furnace and fitted a line over the range that actually matters [3]:

$$\Delta G^\circ = 38,000 - 32.4T \text{ cal mol}^{-1}, \quad T = \frac{38,000}{32.4 - R \ln p_{\text{CO}_2}}$$

At one atmosphere of carbon dioxide the logarithm vanishes, and the temperature comes out at 1,173 kelvin — 899.7 degrees Celsius, which is the handbook number to within a fraction of a degree. It is worth pausing on that, because it is a sixty-year-old measurement recovered from two coefficients.

The second use of the equation is the one the industry actually exploits. The temperature at which limestone decomposes is not a property of limestone; it is the temperature at which the ambient partial pressure of carbon dioxide stops being enough to hold the crystal together. Dilute the gas and the rock gives up sooner. In a modern precalciner the kiln atmosphere runs at roughly a quarter of an atmosphere of carbon dioxide, and the same equation puts the decomposition temperature at 1,081

kELVIN, or 808 degrees — ninety degrees of thermal relief bought purely by sweeping the product gas away. The kinetics are a separate matter, and the decomposition rate depends on the departure from equilibrium rather than on the temperature alone [17], but the equilibrium sets the floor.

The energy budget closes just as neatly. At 179 kilojoules per mole, calcining the lime in a tonne of clinker costs 2.06 gigajoules. The figure quoted for the theoretical heat of clinker formation is 1.75, and the difference is real: the subsequent formation of alite, belite and the aluminate phases is exothermic and hands back about a third of a gigajoule. So of the 3.4 gigajoules a good kiln actually consumes, some 60 per cent is spent pulling carbon dioxide out of rock, and roughly half the total is lost to the second law and to the walls.

The Same Reaction, Running Backwards

Because calcination is reversible, the product wants to go home. Set hardened cement paste in ordinary air and carbon dioxide dissolves in its pore water, attacks the calcium hydroxide left over from hydration and then the calcium silicate hydrate itself, and precipitates calcite. This is carbonation, and it is thermodynamically downhill by the same 179 kilojoules per mole that the kiln had to supply. Nothing has to be built or subsidised to make it happen. The only question is how fast.

The answer has a shape that anyone who has studied diffusion will recognise. Carbon dioxide has to cross the already-carbonated skin before it can reach fresh material, and that skin gets thicker as the front advances. Let x be the depth of the front, D the effective diffusivity of carbon dioxide through the carbonated layer, C_s the concentration at the surface, and a the amount of carbon dioxide the concrete can bind per unit volume. In quasi-steady state the flux arriving at the front is DC_s/x , and all of it is consumed there:

$$a \frac{dx}{dt} = \frac{DC_s}{x} \quad \Rightarrow \quad x = \sqrt{\frac{2DC_s}{a}} \sqrt{t} = k\sqrt{t}$$

Square-root-of-time laws are the signature of a diffusion-limited front, and field measurements on real structures have been fitted to this one for half a century, with k ranging from about 1 to 5 millimetres per root-year depending on the mix, the curing and the weather [12], [13].

What makes it satisfying is that k can be computed rather than fitted. Take ordinary structural concrete at 300 kilograms of cement per cubic metre, a world-average clinker ratio of 0.71 and 65 per cent lime in the clinker: that is 138.5 kilograms of calcined lime per cubic metre, or 2,469 moles, of which perhaps three-quarters is accessible to carbonation, giving $a \approx 1,850$ moles per cubic metre. The atmosphere supplies C_s : at about 425 parts per million and room temperature that is 0.0177 moles per cubic metre [15]. Effective diffusivities for carbonated concrete cluster around 10^{-8} square metres per second. Put the three together and k comes out at 2.45 millimetres per root-year, squarely inside the measured range, with no fitted parameter anywhere in the calculation. The sink is not an empirical curiosity; it is Fick's law with the stoichiometry of a kiln on the other side.

One Cubic Metre, Followed to the End

Now follow a single cubic metre of that concrete. Its 213 kilograms of clinker released 110.8 kilograms of carbon dioxide at the kiln. Suppose it is cast as a 200-millimetre wall exposed on both faces, with $k = 2.5$.

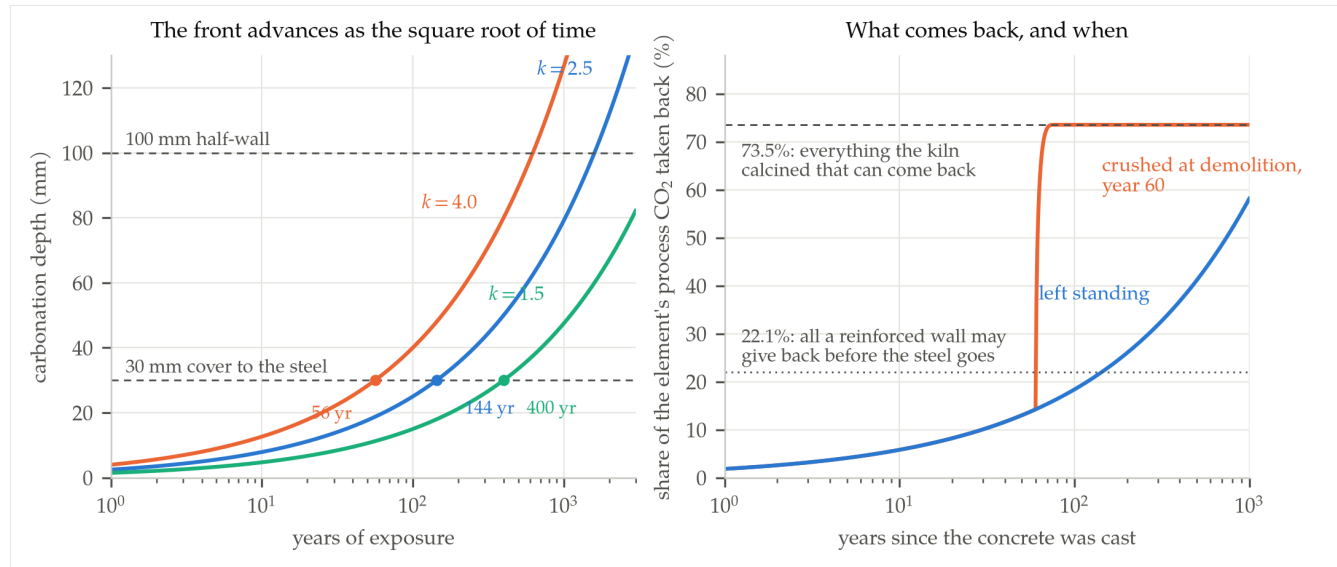


FIGURE Carbonation depth against time for three concrete qualities, and the share of an element's own process carbon dioxide that comes back — for a wall left standing, and for the same concrete crushed at demolition.

There is a hard ceiling before any kinetics are considered. The most the element can ever reabsorb is the carbon that came out of its own clinker, reduced by the fraction of lime that is actually carbonatable and by the kiln dust that left the plant and never became part of the wall. Those two factors are 0.75 and 1.02, so the ceiling is 73.5 per cent of the process emission, or 81.5 kilograms. No exposure, no century, no crushing yard can beat that. Then the square-root law decides how much of the ceiling is reached, and when.

Fate of one cubic metre	Volume carbonated	CO2 back (kg)	Share of its process CO2
Standing 50 years	17.7%	14.4	13.0%
Standing 100 years	25.0%	20.4	18.4%
Front reaches the steel, year 144	30.0%	24.4	22.1%
Standing 1,600 years	100%	81.5	73.5%
Crushed to 20 mm, five years later	91.4%	74.5	67.2%

BEFORE READING ON

The front reaches 30 millimetres — a typical cover to the reinforcement — in 144 years. The wall is 200 millimetres thick. When does the front reach the middle?

Sixteen hundred years, because the depth goes as the root of time and the time therefore goes as the square of the depth: three and a third times deeper takes eleven times longer. That single ratio is why a standing structure is such a poor carbon sink. But the more interesting line in the table is the one at year 144, because it is not a chemical limit at all. Fresh concrete has a pore solution at pH 13, and at that alkalinity steel carries a passive oxide film that simply does not corrode. Carbonation consumes the alkalinity: as the front passes, pH falls below 9, the film is no longer stable, and the reinforcement begins to rust [14]. Tuutti's model of service life — an initiation period while the front travels to the steel, then a propagation period while the steel corrodes and the cover spalls — has organised concrete durability engineering since 1982, and the initiation period is precisely the interval over which a reinforced element is allowed to absorb carbon dioxide.

So for reinforced concrete the two processes are the same process. The cover depth that engineers specify to keep structures standing is also, exactly, a cap on the sink: 30 millimetres of cover on each face of a 200-millimetre wall is 30 per cent of the volume, and 22.1 per cent of the carbon. Design codes size that cover to survive the intended service life with margin [16], which means a well-designed structure is one that carbonates as little as possible. The sink and the failure mode are competing for the same material, and the engineer is required to side against the sink.

Demolition changes everything, and the last row of the table shows why. Crush the same cubic metre into 20-millimetre rubble and you have not changed the chemistry or the diffusivity at all — only the distance from any point to a surface. At the identical $k = 2.5$, a 10-millimetre radius is fully carbonated in sixteen years, and 67 per cent of the original emission is back inside the mineral within five. There is no steel left to protect. This is the mechanism behind every large published estimate of the cement sink, and it is worth being clear that it is a waste-management fact rather than a materials fact: it happens only if the rubble is crushed, exposed and left in the air, rather than buried in a landfill or used as compacted fill below the water table.

The Gigatonne Argument

Scaled up, the global accounts have grown steadily more confident and steadily more contested. Xi and colleagues opened the subject in 2016 with a bottom-up inventory of every cement-using material stock since 1930, and found a cumulative uptake of 4.5 billion tonnes of carbon — about 16.5 billion tonnes of carbon dioxide, offsetting 43 per cent of the process emissions over the same period, with an annual flux that had grown from 0.10 to 0.25 billion tonnes of carbon between 1998 and 2013 [6]. Guo and colleagues extended the series to 2019 and put the cumulative figure at 21.0 billion tonnes of carbon dioxide [7]; the most recent update, built on 58,517 activity-level records from 163 countries, gives 21.26 billion tonnes against 46.06 billion tonnes of cumulative process emissions, or 46 per cent, with an

annual uptake of 0.84 billion tonnes in 2023 [8]. The Global Carbon Budget has adopted the result, and now subtracts a cement carbonation line of about 0.2 billion tonnes of carbon a year from global fossil emissions before reporting them [11]. Against cement process emissions of roughly 1.5 billion tonnes of carbon dioxide a year [1], the sink is not a rounding error.

The dissent has two independent parts. The first is about timing. Van Roijen and colleagues took the same uptake curves and asked what they do to cumulative radiative forcing rather than to a mass balance, and found that because the emission is instantaneous and the uptake is spread over decades, the climate benefit is roughly 60 per cent smaller than the tonnage suggests [9]. A tonne returned in 2090 does not undo a tonne released in 2026; it only stops the forcing accumulating from 2090 onward.

The second is about what is being counted. A 2026 analysis by Xiao, Prentice and colleagues applied thermodynamic and diffusion modelling inside a Monte Carlo framework across real concrete formulations and real element geometries, and concluded that ambient carbonation of concrete in service will take up about 0.23 billion tonnes a year by 2030 — under 10 per cent of the industry's annual emissions, against earlier figures as high as 57 per cent [10]. The cubic-metre calculation above explains the gap without any need to adjudicate: a wall that recovers 13 per cent of its carbon in fifty years cannot produce a sector-wide offset of half.

The reconciliation is in the composition of the earlier estimates, and it is the genuinely surprising part of the subject. In Xi's accounting the cumulative sink breaks down as 58.5 per cent mortar, 30.1 per cent concrete, 7.1 per cent cement kiln dust and 4.0 per cent construction waste. The largest single contributor to the concrete carbon sink is not concrete. It is mortar — the render on a wall, the joint between two bricks, the screed under a floor — which is thin, unreinforced, made at high water content, and assumed in these inventories to carbonate almost completely by the end of its life. Nothing about mortar is constrained by cover depth, because there is no steel in it to protect. The two literatures have been describing different materials.

What the Ceiling Leaves

None of this makes carbonation irrelevant; it makes it a slow, partial and end-of-life phenomenon that cannot be the plan. The plan has to attack the 0.7848 that sits at the top of this piece, and there are only four ways to do it. Use less clinker per tonne of cement, which the world has already done — the clinker ratio fell from about 0.83 in 1990 to 0.69 by 2017 [1] — and can push further with calcined clays and limestone fillers [18]. Use less cement per unit of structure, which is an engineering and procurement problem rather than a chemical one. Capture the carbon dioxide at the plant, which is now demonstrated rather than hypothetical: the Brevik works in Norway began operating the first industrial-scale cement capture plant in June 2025, taking 400,000 tonnes a year, about half the site's emissions, to storage under the North Sea [19]. Or avoid the calcination entirely by sourcing calcium from something that is not a carbonate, which is the bet behind the electrochemical routes now moving out of the laboratory [4].

The lime cycle that a Roman mason understood is still running, and it still closes. What changed is the arithmetic of the two halves. The kiln takes about a second of residence time to send half a tonne of carbon dioxide up a stack, and the wall it eventually becomes takes sixteen centuries to take it back — and is under instruction from its own design code not to try.

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